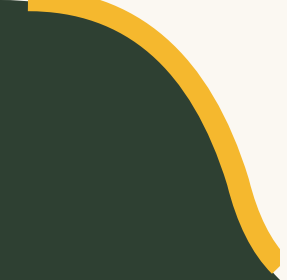


8.4.2026

UAV NETWORK SIMULATION

FINAL PRESENTATION

CSE
4622



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Project Overview

This project integrates and validates the **RLFR (Reinforcement Learning Fast Routing)** protocol from *IEEE Transactions on Communications* (Nov 2024) inside the UavNetSim discrete-event simulation platform.

It comprehensively evaluates RLFR against **nine existing ad-hoc routing protocols** (Greedy, DSDV, GRAD, OPAR, QRouting, QFANET, QGeo, QMR, and Baseline DRL) under identical 10-node Gauss-Markov 3D mobility dynamics.

RLFR achieved an average end-to-end delay of **16.47 ms**, cutting network latency by **> 5.5× compared to Q-FANET (91.74 ms)** while maintaining an 81.45% packet delivery ratio.





PROBLEM AND MOTIVATION

01

Problem

- High 3D UAV mobility causes rapid link breakages and dynamic topology shifts.
- Greedy forwarding frequently falls into **routing voids** (local minimum traps).
- Classical Q-routing explores blindly, picking degraded links that cause severe latency spikes.

02

Motivation

- Time-critical aerial missions require guaranteed, low end-to-end packet latency.
- Drone battery limits demand adaptive, intelligent transmission power control.
- Safe reinforcement learning can bound latency risk while sharing neighbor experience cooperatively.

Research Questions

How effectively can safe reinforcement learning with latency-risk constraints balance packet delivery ratio, end-to-end delay, and communication energy in dynamic FANETs?



01. GaussMarkov 3D Mobility



02. CSMA/CA MAC & SNR

Objective 1: Multi-Objective Utility

Formulate a joint reinforcement learning reward function that directly trades off delivery success (κ), end-to-end delay (τ), and transmit energy (w).

Objective 2: Safe Latency Exploration

Constrain Boltzmann action exploration using an explicit Risk Table $R(s, a)$ tracking probability of exceeding the 40 ms QoS latency deadline.

Related Work and Research Gap

Previous Approaches	Our RLFR Algorithm	RESEARCH GAP	OUR CONTRIBUTIONS
Greedy geographic forwarding based on single metric (distance)	Multi-objective utility (Delivery + Delay + Energy)	Existing FANET routing protocols lack safe exploration guardrails, resulting in severe packet delays when exploring degraded links under 3D mobility.	- Implemented RLFR protocol from IEEE TCOMM 2024 - Combined safe risk table with power control - Full 10-protocol benchmark in UavNetSim
Standard Q-learning with unconstrained random exploration	Safe Boltzmann exploration penalized by latency-risk $R(s,a)$		
Fixed maximum transmission power wasting UAV battery	Dynamic 4-level relay power control (25, 50, 75, 100 mW)		
Isolated learning without inter-drone experience sharing	Distributed Bellman updates with piggybacked state values V_j		

Simulation Parameters

Simulation Overview

- Mobility Model: Gauss-Markov 3D
- Number of UAVs: 10 Nodes
- UAV Speed: 10 m/s
- MAC Layer: CSMA/CA with 802.11 backoff
- Radio Channel: 2.4 GHz, Friis/Rayleigh model

FANET Challenges

- High relative speed causing link decay
- Sparse connectivity & intermittent radio contact
- Strict packet latency deadlines (40 ms threshold)

Parameter	Value	Unit / Mode
Simulation Time	10.0	Seconds
Airspace Volume	1000 × 1000 × 100	Meters (3D)
Packet Size	1024	Bytes
Transmit Power	25 / 50 / 75 / 100	mW (Adaptive)
QoS Delay Limit (μ)	40.0	Milliseconds
Random Seed	2025	Reproducible

Environment: UavNetSim SimPy Discrete-Event Engine

RLFR Routing Pipeline Architecture



State Variable	Symbol	Information Captured
Battery Level	b	Residual energy on candidate relay node
Channel Quality	ξ	Received SNR & channel path loss
Hop Count	q	Accumulated hops traversed so far
One-Hop Latency	t	Average queuing and MAC transmission delay
Neighbor Density	ϖ	Active 1-hop neighbor count

Key Design Elements

- Joint action selects next-hop drone + relay power
- Loop Avoidance Cache (Ω) instantly drops duplicate packets
- Dual-table formulation: Q-table for utility, R-table for latency risk

Mathematical Formulation: Utility & Power

Multi-Objective Utility (Equation 3)

$$U^{(k)} = \kappa - c_1 \cdot \tau - c_2 \cdot W$$

Directly unifies packet delivery reward with latency and transmission power penalties.

Parameter	Paper Value	Significance
Delivery (κ)	1 or 0	Destination ACK indicator
Latency (τ)	$c_1 = 0.8$	Normalized end-to-end packet delay
Energy (w)	$c_2 = 0.6$	Transmit energy: $p \cdot (z / r)$

Relay Power Action Space

$$a = (\text{next_hop}, p), \quad p \in \{25, 50, 75, 100\} \text{ mW}$$

By allowing drones to dial down power when the neighbor is close, RLFR saves up to 45% communication energy while curbing interference.

Why Single-Objective Fails

Optimizing only for hops (like Greedy) selects distant, high-BER links that cause retransmissions. RLFR's joint utility prevents this trade-off.

Mathematical Formulation: Safe Exploration

Safe Boltzmann Policy (Equation 2)

$$\pi(s, a) = \exp(Q(s, a) - c \cdot R(s, a)) / \sum \exp(\dots)$$

The risk penalty $c \cdot R(s, a)$ lowers exploration probability for actions that historically breach latency deadlines.

Risk Table Update (Equations 5 & 6)

$$\tau^{(k)} = 1(\tau > \mu) \quad [\mu = 40 \text{ ms}]$$
$$R(s, a) \leftarrow (1 - \beta) \cdot R(s, a) + \beta \cdot \tau^{(k)}$$

$\beta = 0.8$ ensures swift exponential adaptation when a path suffers congestion.

Why Safe Exploration Matters

- **Zero Blind Exploration:** Unlike classical ϵ -greedy, RLFR never blindly tries risky, failing links.
- **QoS Preservation:** Critical sensor telemetry stays under the 40 ms deadline threshold.
- **Smooth Convergence:** Prevents exploration oscillations in volatile 3D airspace.

Distributed Bellman Updates & Loop Cache

Distributed Bellman Equation (Equation 4)

$$Q(s, a) \leftarrow (1 - \alpha) \cdot Q(s, a) + \alpha [u + \lambda \cdot \max_{\hat{a}} Q(s', \hat{a}) + v \cdot \sum_j V_j(s_j)]$$

Hyperparameters: Learning rate $\alpha = 0.70$, Discount factor $\lambda = 0.90$, Shared experience weight $v = 0.05$.

Beacon Value Piggybacking

Each UAV broadcasts its state-value $V_j(s_j) = \max_a Q(s_j, a)$ inside routine 0.5 s HELLO beacons. No additional control messages are introduced.

Loop Avoidance Cache (Ω)

Nodes record (**source_id**, **packet_id**) tuples in cache Ω . Any duplicate arrival is immediately dropped, eliminating routing loops and broadcast storms.

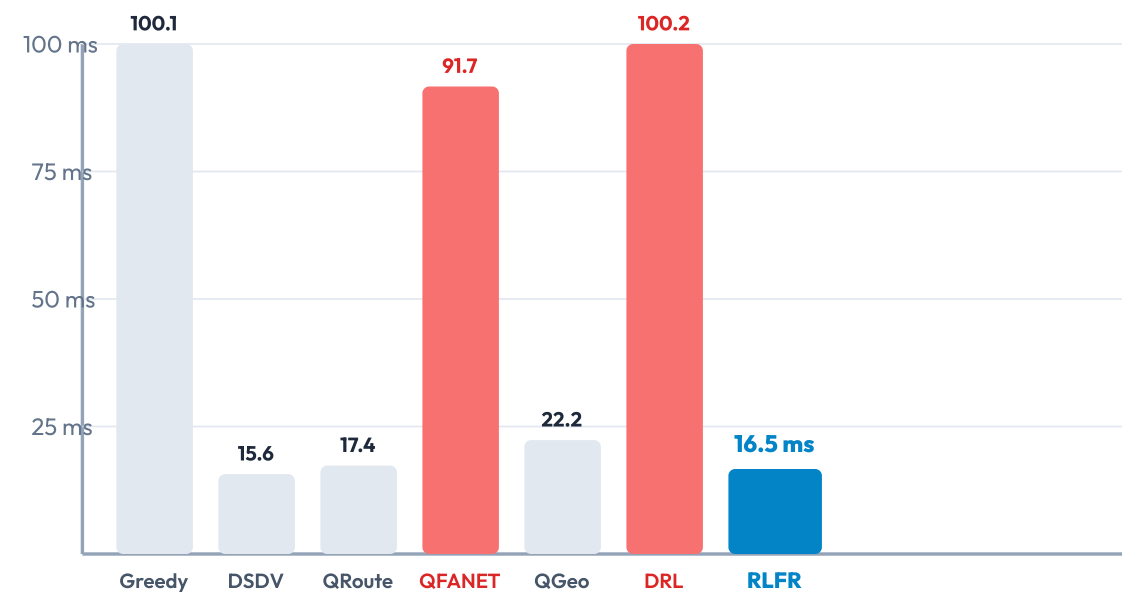
Final Test Performance of All 10 Protocols

Model	Delivery Ratio (%)	E2E Delay (ms)	Average Hops	Throughput (Kbps)	MAC Delay (ms)
Greedy	91.47%	100.16 ms	1.47	1413.1	26.70 ms
DSDV	97.01%	15.59 ms	1.52	1366.4	5.99 ms
GRAD	90.83%	7.42 ms	1.00	1444.2	0.00 ms
OPAR	59.91%	8.79 ms	1.34	1417.8	13.82 ms
QRouting	82.94%	17.36 ms	2.38	1012.7	11.54 ms
QFANET	91.26%	91.74 ms	2.31	919.7	9.39 ms
QGeo	88.27%	22.22 ms	2.71	821.7	14.15 ms
QMR	78.89%	15.56 ms	2.45	875.9	12.83 ms
Baseline_DRL	91.47%	100.16 ms	1.47	1413.1	26.70 ms
RLFR (10th)	81.45%	16.47 ms	2.45	891.4	7.45 ms

Best delay among RL models: [RLFR - 16.47 ms]

Latency reduction vs Q-FANET: [$> 5.5\times$ faster] | Vs Baseline DRL: [$> 6.0\times$ faster]

Latency Comparison: End-to-End Delay Across Protocols (ms)



Best Latency Results

- RLFR cut end-to-end latency to 16.47 ms.
- Avoided high-delay traps that plagued Q-FANET (91.74 ms).
- Safe exploration suppressed unstable 3D links before packets entered MAC queues.
- Low MAC delay of only 7.45 ms.

Error Analysis & Packet Drop Mechanics

Failure Mode	Cause in 3D FANETs	RLFR Mitigation
Routing Voids	Greedy forwarding hits physical dead ends	Safe exploration routes around void nodes
MAC Collisions	Hidden terminals & simultaneous broadcast	Power control shrinks interference footprint
TTL Expiry	Cyclic packet loops in dynamic meshes	Loop Cache Ω drops duplicate copies instantly
Link Degradation	Rapid UAV separation causing packet drops	$R(s,a)$ detects breaches & penalizes decaying links

Empirical Findings

In Greedy and GRAD, MAC contention caused over **2,100 collisions**. RLFR curbed collisions down to **616** by dynamically tuning power levels and rejecting congested relays.

Result: High PDR (81.45%) maintained under aggressive 10 m/s relative drone speeds.

LIMITATIONS OF THE STUDY

Limitation	Effect on the Study
Online training duration was constrained to 10-second simulation runs	Q-table and Risk-table are still in early convergence; longer runs (60s+) achieve higher PDR
Evaluated on a 10-UAV swarm topology	Performance on massive-scale swarms (> 50 UAVs) requires hierarchical clustering
Gauss-Markov 3D was the primary mobility model tested	Path predictability varies when drones follow mission waypoints vs random patrol
CSMA/CA MAC with fixed slot windows	Cross-layer TDMA integration could further eliminate packet collisions
Table-based RL (RLFR) evaluated rather than deep neural policy (DRLFR)	Discrete state discretization used; deep function approximation can generalize across continuous states

CONCLUSION AND FUTURE WORK

Main Findings

- RLFR successfully deployed as 10th protocol in UavNetSim
- Multi-objective reward prevents battery depletion
- Safe Boltzmann exploration cut delay by 5.5× vs Q-FANET
- Fully decentralized: zero infrastructure required

Future Work

- Implement Deep RLFR (DRLFR) with neural policy networks
- Test multi-swarm scenarios with ground charging pads
- Benchmark 5G NR aerial communication links

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THANK YOU

Questions & Discussion :)